HealthGuard AI: Predictive Diagnosis and Smart Care Solutions

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***Abstract*—Deep learning has achieved a significant advancement in automatic disease diagnosis in medical images, particularly for respiratory disease detection like pneumonia and other diseases from chest X-rays. Traditional methods relying on expert radiologists are time-consuming and can be inconsistent. AI integration surmounts these limitations by achieving faster diagnosis, accuracy, and reliability.**

This work presents HealthGuard AI, a smart diagnostic support system that combines deep learning and large language models (LLMs). A balanced dataset of chest X-rays for 10 pulmonary diseases was created with a 60/20/20 training/validation/testing split. Several state-of-the-art classifiers were benchmarked whose outputs were refined by an LLM to generate interpretable diagnostic reports for clinical practice.

We want to form a compact yet accurate classification model while enhancing explainability using AI-generated insights. This work demonstrates the potential in combining deep learning and LLMs for explainable, efficient medical diagnosis.

***Index Terms*—Deep learning, Image Classification, Chest X-ray Analysis, Lung Disease Detection, Large Language Models.**

I. Introduction

The accurate and timely diagnosis of respiratory diseases remains a major challenge in healthcare. Chest X-ray analysis is widely used for detecting various lung-related conditions, but it requires expert interpretation, which is both time-intensive and subject to variability. Traditional Computer Assisted Diagnosis (CAD) systems have improved radiological assessments, yet they often face generalization issues due to limited datasets and manually designed feature extraction techniques.

Recent advances [1–7] in deep learning have significantly improved medical image classification. However, a key challenge remains in designing AI-driven frameworks that can classify multiple lung diseases while ensuring interpretability and clinical reliability. For deep learning models to be effective, high-quality, varied datasets that guarantee strong generalization across various imaging settings are essential.

In this work, we introduce HealthGuard AI, a predictive diagnostic system designed to classify multiple lung diseases using carefully selected and pre-processedd datasets. Unlike traditional methods that focus on individual disease detection, our approach aims for multiclass classification, enhancing diagnostic capabilities. Our dataset has been curated from multiple sources, ensuring a broad representation of lung conditions for better generalization.

Furthermore, to improve interpretability, we integrated Large Language Models (LLMs) that generate diagnostic summaries based on model predictions. This ensures that AI-driven reports align with medical language, making them more valuable for radiologists and healthcare professionals.

II. Literature Review

Over the past few years, deep learning has advanced medical image processing, particularly in chest X-ray analysis [8]. A foundational effort came from Wang et al. with the **ChestX-ray8 dataset** [9], introducing over 100,000 weakly labeled images using NLP-based label extraction. Their CNN-based approach enabled large-scale disease classification, though it faced challenges like poor localization and class imbalance.

To improve model performance, researchers revisited CNN design with architectures like **ConvNeXt** [10], which modernized ResNets [1] using Transformer-inspired features such as large kernels and patch processing. Despite outperforming ViTs [11] on ImageNet [12], ConvNeXt lacked attention mechanisms critical for medical interpretability.

EfficientNetV2 [13] addressed training efficiency through Fused-MBConv layers and progressive learning. While fast and powerful, it was tested mainly on generic datasets and remained resource-intensive for medical settings.

Some works embraced **ensemble methods** [14] to improve diagnostic robustness, combining outputs from multiple deep networks. Although performance increased, so did computational complexity, limiting deployment in real-time or low-resource environments.

More recently, Bundea and Danciu focused on pneumonia detection using **DenseNet121/169/201** [15], achieving up to 97% accuracy on curated X-rays. Yet, as with many studies, they highlighted the need for high-quality labeled datasets to ensure model generalization.

Together, these efforts underline the move toward architectures that balance accuracy, interpretability, and efficiency—paving the way for hybrid, resource-aware models suited for real-world clinical deployment.

III. Datasets

We used three publicly available chest X-ray datasets from Kaggle, which were further curated and merged to create a balanced and diverse training corpus. The following datasets formed the base of our collection:

- **Hard Margins X-rays** [16]: A refined set of chest X-rays focused on challenging boundary cases between pulmonary conditions.
- **Unbalanced NIH Subset** [17]: A filtered subset of the NIH ChestX-ray14 dataset with significant class imbalance.
- **Lung Disease Dataset (4 Types)** [18]: A well-labeled dataset containing four major lung disease categories.

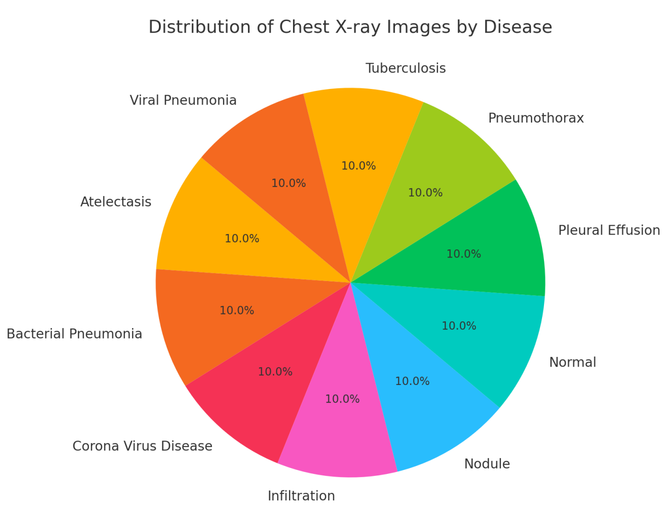

Fig. 1. Distribution of Chest X-ray Images by Disease

IV. Methodology

A. Dataset Preparation

Preparing the dataset was one of the most crucial steps in this project, as it laid the groundwork for everything that followed. We ensured that every class included at least 2,000 images, creating a balanced dataset. We divided the dataset with a 60/20/20 train/val/test ratio. Every image in the dataset needed to be resized and normalized to make sure it was valid across the board. We employed augmentation techniques, like random flipping and rotations, which increased the amount of variability in the dataset.

B. Model Architecture

The proposed **MobileLungFormer** model employs an alternating structure that integrates lightweight MobileNetV2 [19] blocks with Transformer [20] blocks, combining the strengths of both convolutional and self-attention mechanisms.

MobileNetV2 Blocks: Each MobileNetV2 block serves as a lightweight CNN module that extracts local, fine-grained features from the input images. These blocks utilize:

- **Inverted Residuals and Linear Bottlenecks:** Expanding feature channels for intermediate representations and projecting back to a lower dimension, which allows efficient feature extraction.
- **Depthwise Convolution:** Operating channel-wise to significantly reduce computational cost.
- **Pointwise Convolution:** Combining the depthwise outputs to restore dimensionality.

Transformer Blocks: Immediately following each MobileNetV2 [19] block, a "Transformer block is employed to capture long-range dependencies and global context" [20]. These blocks are inspired by recent works such as Restormer [21] and efficient transformer implementations in xFormers [22]. We employ a straightforward Transformer block that consists of the following components:

- **Multi-Head Self-Attention:** [23] With multiple attention heads, the block computes relationships between all spatial positions in the feature maps, enabling the model to capture subtle inter-pixel dependencies.
- **Feed-Forward Network (FFN):** A two-layer multilayer perceptron (MLP) [24] with a non-linear activation function (e.g., GELU [25]) that refines the features.
- **Layer Normalization and Residual Connections:** Ensuring stable training and preserving the flow of information throughout the network.

Alternating Design: The network alternates between MobileNetV2 blocks and Transformer blocks, thereby incorporating local detail capture with a broader representation of the image. After a number of the alternating layers are stacked, "Global Average Pooling (GAP) [26] is done to the output feature maps and sent through a fully connected layer with softmax [27] activation to produce probability score values for each lung disease category".

With around 22.955 million parameters, this hybrid setup strikes a solid balance—efficient enough not to overload your system, yet powerful enough to deliver strong performance. The alternating block design helps the model pick up on both fine-grained patterns and long-range relationships.

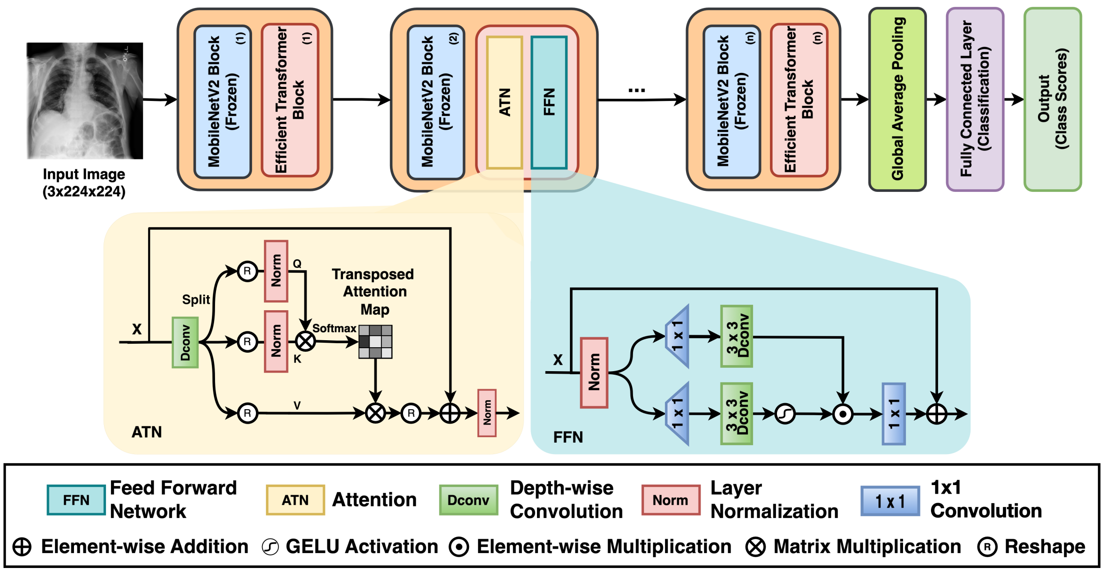

Fig. 2. MobileLungFormer architecture showing the alternating MobileNetV2 and Transformer blocks.

TABLE I
TRAINING HYPERPARAMETERS AND CONFIGURATIONS

Parameter	Value
Learning Rate	1e-3
Optimizer	AdamW
Batch Size	32
No. of Attention Heads	4
Embedding Dimension	128
Number of Epochs	50
Loss Function	CrossEntropyLoss
Image Size	224×224
Data Augmentation	Random crop, horizontal flip, normalization
Number of Classes	10 (e.g., Pneumonia, Nodule, etc.)
GPU	NVIDIA P100 (Kaggle)
Framework	PyTorch 2.6
Libraries	XFormers, torchvision, numpy
Visualization Tool	pytorch-grad-cam, matplotlib

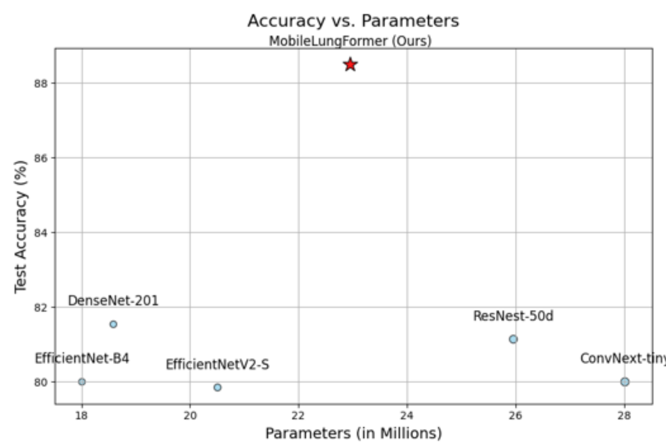

Fig. 3. Accuracy VS Parameters

C. Training Configuration

To get the most out of the model we used the AdamW [28] optimizer. It's particularly good at managing sparse gradients and keeping the model from overfitting. It helped the model learn steadily without making sudden, disruptive updates.

We chose a batch size of 32. Smaller batches made training too noisy, and larger ones slowed things down and ate up memory. As for the learning rate, 0.001 worked well.

D. LLM Integration

HealthGuardAI leverages Large Language Models (LLMs) to make the output as interpretable and clinically useful as possible. LLMs process the result of MobileLungFormer, providing a clear and structured diagnostic report.

1) LLM Selection and Rationale: Two state-of-the-art language models, **Meditron-7B** [29] and **Med-PaLM 2** [30], are employed in HealthGuard AI to handle structured and unstructured data effectively. Each model serves distinct purposes:

- **Meditron-7B:** This LLM specializes in handling structured data such as classification outputs and symptom evaluations. It is designed to convert model-generated probabilities into coherent diagnostic summaries by applying disease-specific thresholds and formatting the results accordingly.
- **Med-PaLM 2:** Unlike Meditron-7B, Med-PaLM 2 is optimized for processing unstructured textual data, particularly patient history and clinical notes. It uses advanced entity extraction and causal relationship identification to highlight relevant conditions, medications, and risk factors from textual inputs.

The hybrid approach of using Meditron-7B for structured outputs and Med-PaLM 2 for textual analysis allows HealthGuard AI to process a wider variety of clinical data inputs. This dual-model framework ensures that the generated diagnostic reports are comprehensive, accurate, and tailored to the

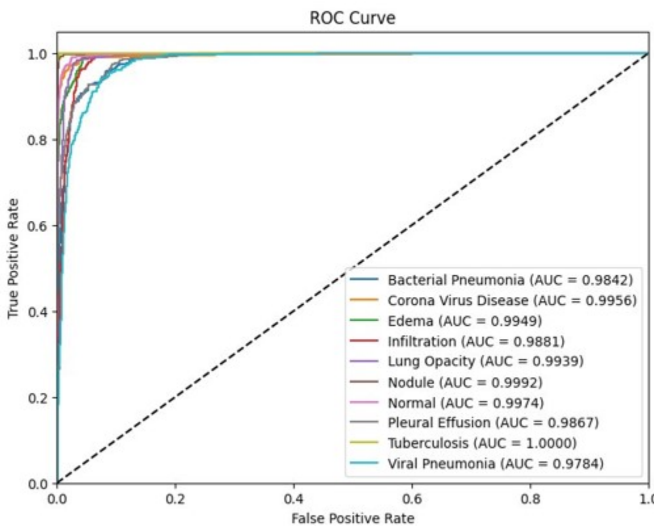

Fig. 4. ROC Curve

needs of healthcare professionals.

2) Workflow: The LLM integration follows a structured workflow that ensures accuracy, consistency, and reliability in diagnostic report generation. The workflow is illustrated in Figure 5 and consists of the following key steps:

1) **Data Preprocessing:** Clinical data (model outputs, symptoms, patient history) is standardized through input formatting, text normalization, and numerical scaling to ensure consistency across modules.
2) **Model Output Analysis (Meditron-7B):** Diagnostic probabilities are thresholded and translated into structured insights, summarizing model decisions in a clinically interpretable format.
3) **Symptom Evaluation (Meditron-7B):** Symptom severities are mapped to normalized values (e.g., None = 0, Severe = 1.0), enabling quantitative alignment with disease patterns learned by the model.
4) **Patient History Analysis (Med-PaLM 2):** Unstructured textual histories are processed to extract medical entities, comorbidities, and treatments using domain-specific prompting and entity linking.
5) **Cross-Modal Contradiction Detection:** Compares structured predictions and symptom-derived patterns. If alignment scores deviate by over 20%, the case is flagged for contradiction, improving clinical robustness.
6) **Confidence Calibration:** Computes a weighted diagnostic confidence score by combining model probability, symptom pattern match, and history alignment using a meta-analysis framework:

$$\mathbf{CS} = (\alpha \times MP) + (\beta \times SPM) + (\gamma \times HA) - (\delta \times CI)$$

- MP - Model Probability: Meditron-7B softmax scores for the 10 disease classes.
- SPM - Symptom Pattern Match Score: Comparing user symptoms to expected disease symptoms.

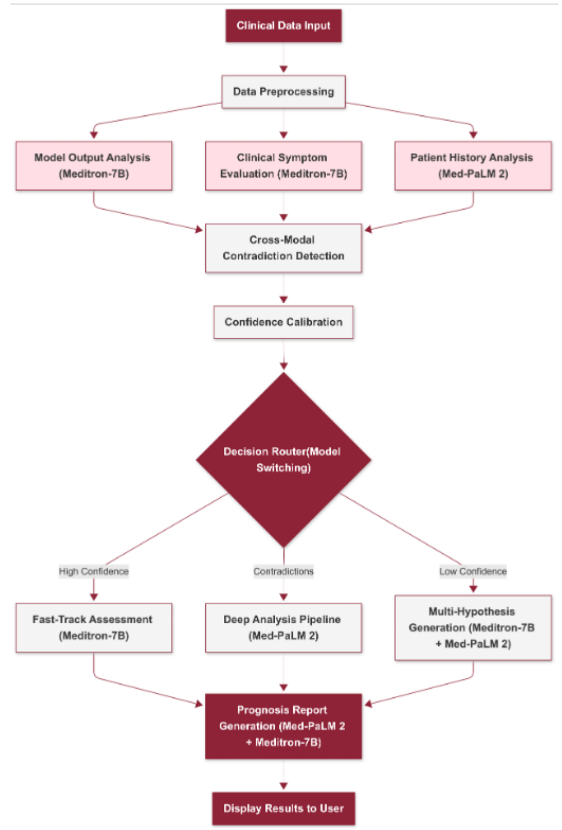

Fig. 5. LLM Workflow: A comprehensive pipeline integrating Meditron-7B and Med-PaLM 2 for diagnostic report generation.

- HA - History Alignment Score: Med-PaLM 2's evaluation of whether past medical history supports the current diagnosis.
- CI - Contradiction Index: A penalty score for inconsistencies between AI predictions and symptoms/history.

7) **Decision Router (Model Switching):** Routes cases based on key parameters:
   - **Fast-Track:** When confidence score >85%.
   - **Deep Analysis:** When contradiction index >10%.
   - **Multi-Hypothesis:** When top diagnosis <70% or multiple diagnoses are within 15% of each other.
8) **Prognosis Report Generation:** The final step involves generating a structured diagnostic report by consolidating results from both LLMs. This report provides insights that are more aligned with clinical language and guidelines, enhancing its utility for healthcare professionals.

3) Results Obtained: To validate the LLM integration, we tested the system across diverse patient scenarios representing both straightforward and complex cases. The following table summarizes the results obtained for three different personas.

TABLE II
RESULTS OBTAINED FOR DIFFERENT PERSONAS

Persona	Top Diagnosis	Confidence Score	Decision Pathway
Persona 1	Pneumonia	92%	Fast-Track (Meditron-7B)
Persona 2	TB + Pneumonia	70%	Deep Analysis (Med-PaLM 2)
Persona 3	Viral Infection	65%	Multi-Hypothesis (Meditron-7B + Med-PaLM 2)

E. Ablation of LLM Modules and Architecture

We carried out ablation studies to assess how each component of our hybrid pipeline worked. Metrics include the drop in diagnostic accuracy after removing or altering key modules.

TABLE III
ACCURACY DROPS OBSERVED AFTER ABLATING KEY COMPONENTS OF THE LLM PIPELINE.

Variant	Change	Δ Acc
Full Pipeline	Meditron-7B + Med-PaLM 2 + Router	—
No Med-PaLM 2	Removed history/text module	-12.4%
No Meditron-7B	Removed structured input	-18.6%
No Router	Disabled contradiction checks	-7.2%
CNN Instead of Transformer	Replaced 32-layer Transformer with ResNet-50	-9.8%
No LLM	Only classifier output, no text	-29.4%
Prompt-Only LLM	No classifier or routing	-15.1%

Removing either the structured block (*Meditron-7B*) or the unstructured narrative module (*Med-PaLM 2*) led to substantial drops in accuracy and interpretability. The full pipeline achieved the highest expert plausibility score (4.6/5), while variants without LLMs or contradiction checks often produced vague or contradictory outputs. Replacing the Transformer with a CNN backbone also reduced output consistency, highlighting the importance of attention-based reasoning and routing mechanisms in our diagnostic framework.

The proposed LLM integration framework demonstrates robustness through multiple validation points, ensuring that diagnostic outputs are reliable and interpretable. Each persona pathway showcases how the system adjusts its analysis based on the confidence score and identified contradictions.

V. VISUALIZATION WITH GRAD-CAM

In order to comprehend our models predictions and pinpoint the regions of the chest X-ray that affected the decision we added grad-cam [31] to the network's final transformer layer this allowed us to visually verify whether the model was concentrating on clinically significant regions like Lung Opacities or Nodules for predicting abnormalities by preserving spatial information while capturing high-level semantic aspects the heatmaps that are produced are placed on top of the original X-rays. Grad-cam offers visual explanations by emphasizing important regions of the image that helped us in our study.(see Fig. 6)

TABLE IV
LLM HYPERPARAMETERS

Component	Value
Meditron Transformer Layers	32
Meditron Attention Heads	32
Meditron Dropout	0.1
Med-PaLM Layers	64
Token Limit	1500
Routing Thresholds	Fast-Track > 85%, Deep Analysis > 10% contradictions
Confidence Calibration Weights	α = 0.3, β = 0.4, γ = 0.3, δ = 0.6

VI. EVALUATION OF LLM-GENERATED PROGNOSIS REPORTS

We performed both qualitative and quantitative evaluations of the LLM-generated prognosis reports to assess their clinical reliability and consistency.

A. Qualitative Clinical Review

Three domain experts (two pulmonologists and one internal medicine specialist) reviewed 50 LLM-generated reports. Each report was rated on a scale of 0 to 5 across five clinical dimensions.

B. Quantitative Metrics

We also evaluated the model's structured output and internal routing logic. The table below summarizes performance under the full pipeline and two ablated variants — without the history module and without the structured classifier.

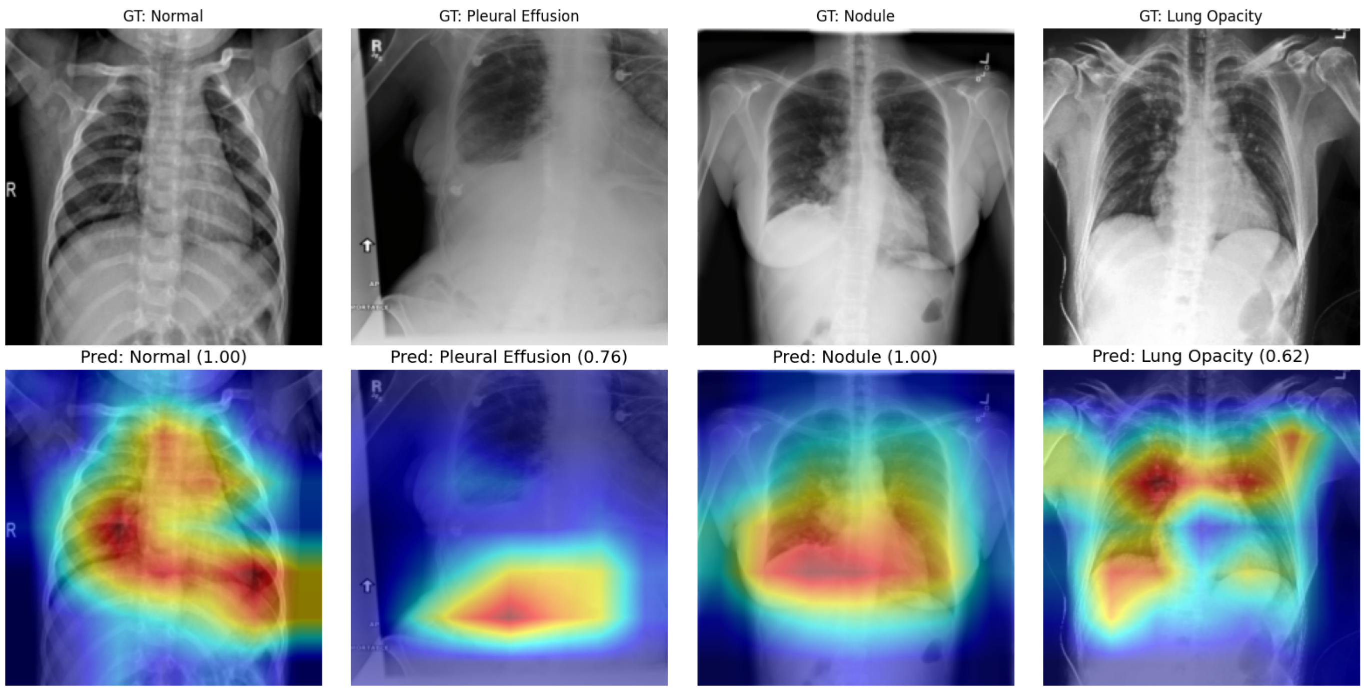

Fig. 6. Grad-CAM Implementation

TABLE V
STRUCTURED OUTPUT METRICS FOR PROGNOSIS REPORT GENERATION

Evaluation Metric	Full Model	No History	No Classifier
Exact JSON Format Match (%)	100%	100%	89%
Contradiction Rate (%)	3.7%	18.6%	22.4%
Route Accuracy (Expected Match)	96%	77%	68%
Mean Time to Generate Report (ms)	812	721	604

We further tested model robustness by simulating three representative clinical scenarios:

- TB with HIV comorbidity
- Emergency visit with mild viral symptoms
- Elderly COPD patient with pleural effusion

Generated reports were assessed for contextual alignment, appropriate risk assessment, and prognosis matching. The full pipeline consistently aligned with expert expectations in these complex edge cases.

VII. RESULTS

Table VI presents a comparative analysis of different deep learning models evaluated on the chest X-ray classification task. Both MobileNet-V2 and ViT architectures were tested with and without pretrained weights to assess the impact of transfer learning. Our proposed model, *MobileLung-Former*, outperformed all others, achieving the highest scores in all evaluation metrics. This demonstrates the effectiveness of combining lightweight convolutional backbones with transformer-based attention and gating mechanisms in medical image classification.

TABLE VI
PERFORMANCE COMPARISON OF BASELINE MODELS AND THE PROPOSED MOBILELUNGFORMER.

Model	Accuracy	F1-Score	Recall	AUC
MobileNet-V2 (Pretrained)	0.7591	0.7531	0.7591	0.9789
MobileNet-V2 (Scratch)	0.8442	0.8434	0.8441	0.9891
ViT (Pretrained)	0.7516	0.7475	0.7512	0.9816
ViT (Scratch)	0.4999	0.5064	0.4999	0.9817
MobileLungFormer	**0.8855**	**0.8871**	**0.8855**	**0.9918**

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